North South University

Department of Electrical & Computer Engineering

Fall 2025

Course Title: Analog Electronics I Lab

Faculty Name: Ms. Syeda Sarita Hassan [SSH1]

Section: 7

Experiment Number:05

Experiment Name:

The Input-Output characteristics of CE (common emitter) configuration of BJT.

Experiment Date: 11/11/25

Date of Submission: 25/11/25

Group Number:01

Submitted To: Md. Mahbub Hassan

Submitted By: Mobussira Tasnim Pushpita-2311032042			Score
Serial	Student Name	ID	
1.	Mobussira Tasnim Pushpita	2311032042	
2.	Tasnuva Islam	2233379642	
3.	Maimuna Islam Muna	2311818042	

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Experiment name: The input-output characteristics of CE (common emitter) configuration of BJT

Objectives:

- To study the input characteristics (I_B vs V_{BE}) of common emitter configuration.
- To study the output characteristics (I_C vs V_{CE}) for different base currents.
- To identify the ~~three~~ operating regions.
- To determine current gain ($\beta = I_C/I_B$) in the active region.
- To understand how input and output characteristics help in selecting a proper Q-point for amplification.

Theory: A transistor is a three-layer semiconductor device consisting of either two n and one p type layers of material or two p and one n type layers of material. A transistor has three doped regions such as:

Emitter (E): The emitter is a segment of one side of transistor of moderate size and its heavily dopped. It emits or inject free majority carrier (electron for NPN and hole for PNP) into the base.

Base (B): The base is the central segment of thin size lightly doped. It passes most of the emitter-injected electron (for NPN) into the collector.

Collector (C): The doping level of collector is between emitter and base.

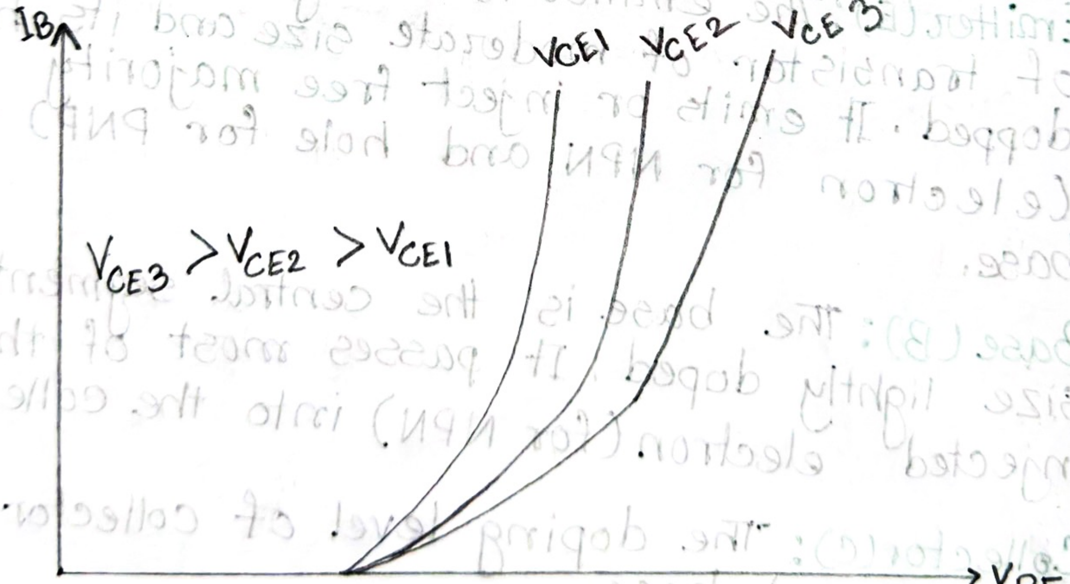
If the V_{BB} is greater than the barrier potential, emitter electron will enter base region. The free electron can flow either into the base or into the collector. As the base is lightly dopped and thin, most of the free electron will enter into the collector. There are three different currents in a transistor. They are emitter current (I_E), collector current (I_c) and the base current (I_B).

Emitter current is the sum of the collector and base currents: $I_E = I_B + I_c$

Current gain, $\beta = I_c / I_B$

Characteristics of a transistor: The characteristics of a transistor is measured by two characteristic curve.

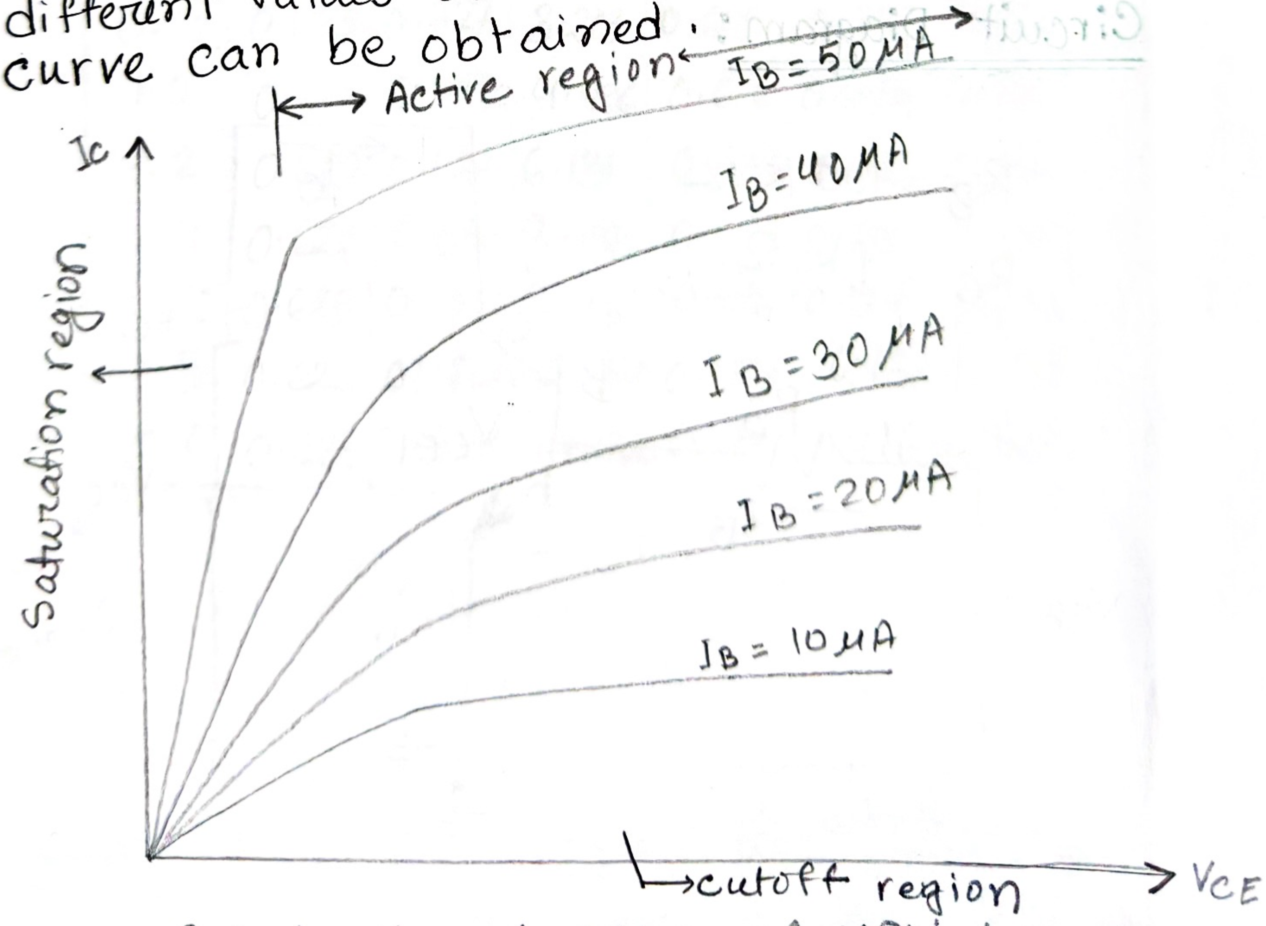
a. Input characteristics curve: Input characteristics is defined as the set of curves between input current (I_B) vs input voltage (V_{BE}) for the constant output voltage (V_{CE}). It is the same curve that is found for a forward biased diode.



Input Characteristic

b. Output characteristics curve: It is defined by the set of curves between output current (I_c) vs output voltage (V_{CE}) for the constant input current (I_B).

Output characteristics curve has three regions. They are Active on, saturation and cutoff region. The rising part of the curve, where V_{CE} is between 0 and approximately 1 volt is called saturation region. In this region, the collector diode is not reversed biased. When the collector diode becomes reverse biased, the graph becomes horizontal. In Active region, the collector remains almost constant. In applications where the transistor amplifies weak radio and TV signal, it will always be operating in active region. When the base current is zero, but there is some collector current. This region of the transistor curve is known as the cutoff region. The small collector current is called collector cutoff current. For the different values of base current (I_B) an individual curve can be obtained:



Output characteristics of NPN transistor

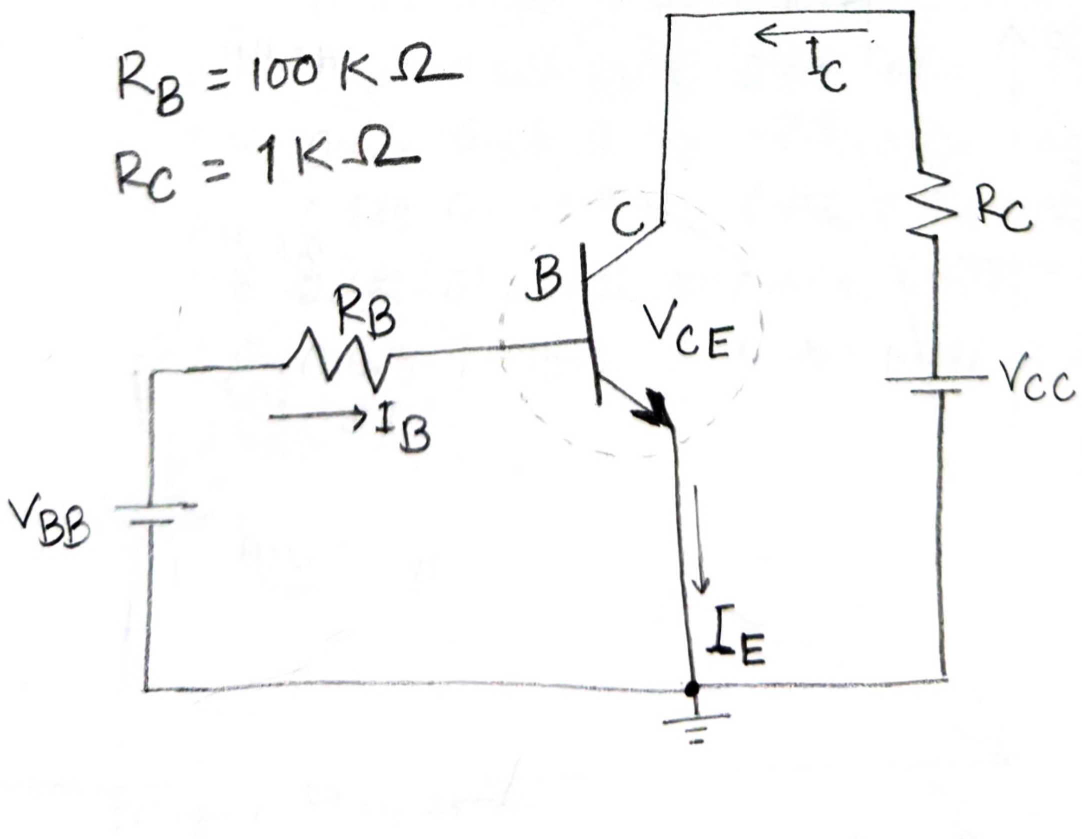
List of Equipment:

Serial	Components Details	Specification	Quantity
1	Transistor	C828	1 piece
2	Resistor	100KΩ, 1KΩ	1 piece each
3	Trainer Board		1 unit
4	DC power supply		1 unit
5	Digital Multimeter		1 unit
6	~~Co~~ Chords		as required
7	Wire		as required

Circuit Diagram:

$R_B = 100\,K\Omega$

$R_C = 1\,K\Omega$



Data Table:

$R_B = 102.1 k\Omega$] Practical values.
$R_C = 0.98 k\Omega$]

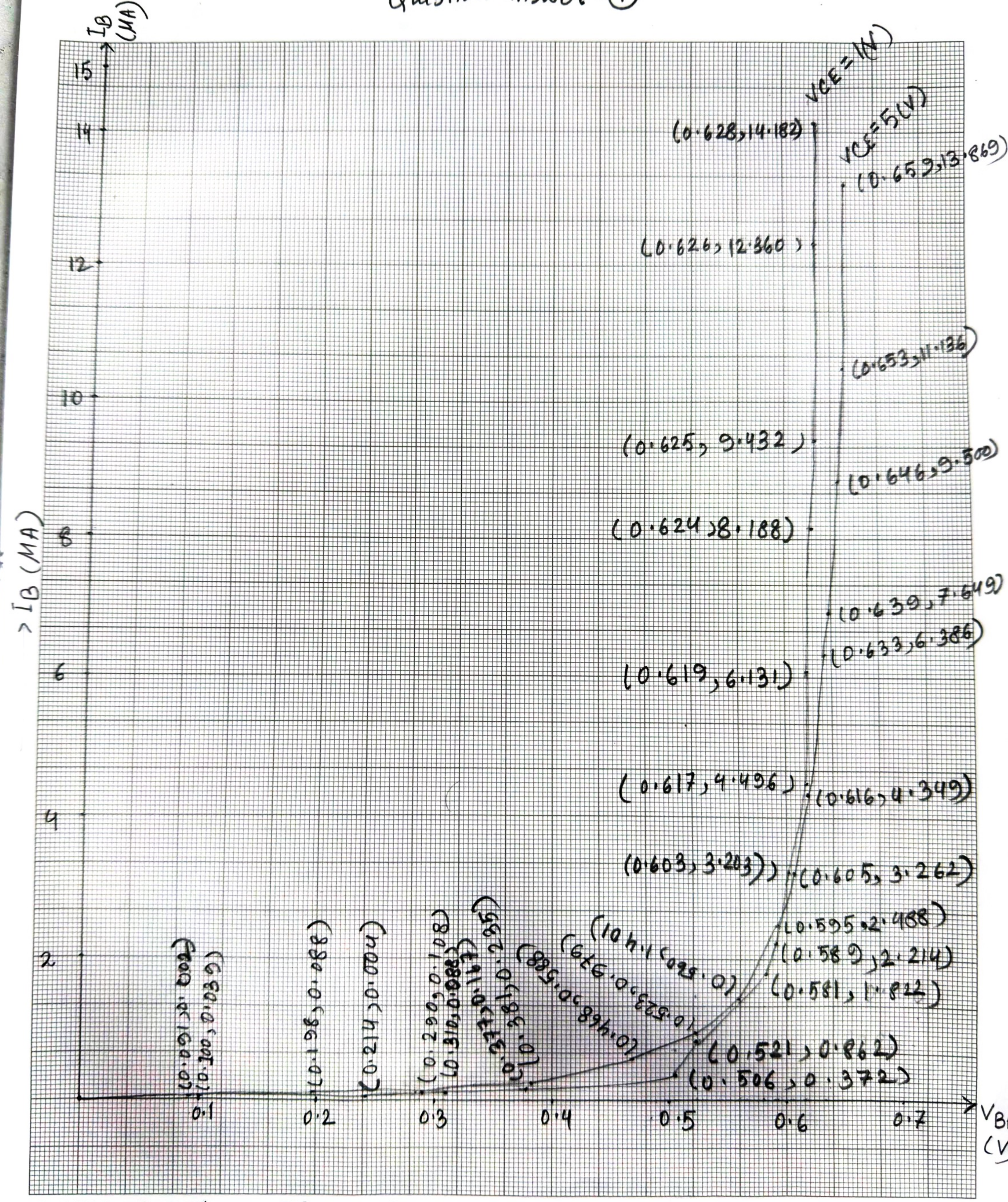
Input Characteristics of BJT

$V_{CE} = 1V$ (0.992V)				$V_{CE} = 5V$ (4.998V)		
V_{BB} (V)	V_{BE} (V)	V_{RB} (V)	$I_B = \frac{V_{RB}}{R_B}$ (μA)	V_{BE} (V)	V_{RB} (V)	$I_B = \frac{V_{RB}}{R_B}$ (μA)
0.1	0.091	0.002	0.020	0.100	0.004	0.039
0.2	0.214	0.004	0.039	0.198	0.009	0.088
0.3	0.310	0.009	0.088	0.290	0.011	0.108
0.4	0.377	0.017	0.167	0.381	0.024	0.235
0.5	0.506	0.038	0.372	0.468	0.060	0.588
0.6	0.521	0.088	0.862	0.523	0.100	0.979
0.7	0.581	0.186	1.822	0.560	0.143	1.401
0.8	0.589	0.226	2.214	0.595	0.254	2.488
0.9	0.603	0.327	3.203	0.605	0.333	3.262
1.0	0.617	0.459	4.496	0.616	0.444	4.349
1.2	0.619	0.626	6.131	0.633	0.652	6.386
1.4	0.624	0.836	8.188	0.639	0.781	7.649
1.6	0.625	0.963	9.432	0.646	0.970	9.500
1.8	0.626	1.262	12.360	0.653	1.137	11.136
2.0	0.628	1.448	14.182	0.659	1.416	13.869

Output Characteristics of BJT

	$I_B = 10\mu A$ ($V_{RB}=1V$)			$I_B = 50\mu A$ ($V_{RB}=5V$)		
V_{CC} (V)	V_{CE} (V)	V_{RC} (V)	$I_C = \frac{V_{RC}}{R_C}$ (mA)	V_{CE} (V)	V_{RC} (V)	$I_C = \frac{V_{RC}}{R_C}$
0.1	0.020	0.070	0.071	0.017	0.110	0.112
0.2	0.035	0.148	0.151	0.029	0.239	0.244
0.3	0.048	0.243	0.248	0.032	0.310	0.316
0.4	0.062	0.392	0.400	0.002	0.362	0.369
0.5	0.067	0.454	0.463	0.003	0.474	0.484
0.6	0.075	0.567	0.579	0.020	0.580	0.592
0.7	0.080	0.662	0.676	0.040	0.690	0.704
0.8	0.084	0.752	0.767	0.050	0.811	0.828
0.9	0.088	0.816	0.833	0.050	0.960	0.979
1.0	0.091	0.885	0.903	0.051	0.971	0.990
1.2	0.100	1.099	1.121	0.052	1.160	~~0~~1.184
1.5	0.113	1.429	1.458	0.060	1.480	1.489
2.0	0.133	1.939	1.979	0.062	~~2.466~~ 1.940	1.979
2.5	0.154	2.385	2.434	0.070	~~2.900~~ 2.966	2.516
3.0	0.203	2.863	2.921	0.080	~~2.921~~ 2.900	2.959
~~3~~5.0	1.907	3.064	3.127	0.110	~~9.870~~ 4.921	5.021
10.0	6.990	3.077	3.139	0.190	~~13.955~~ 9.870	10.071
15.0	11.560	3.500	3.571	1.030	13.955	14.234
20.0	16.080	3.980	4.061	5.120	14.960	15.265

Qustion Answer -1



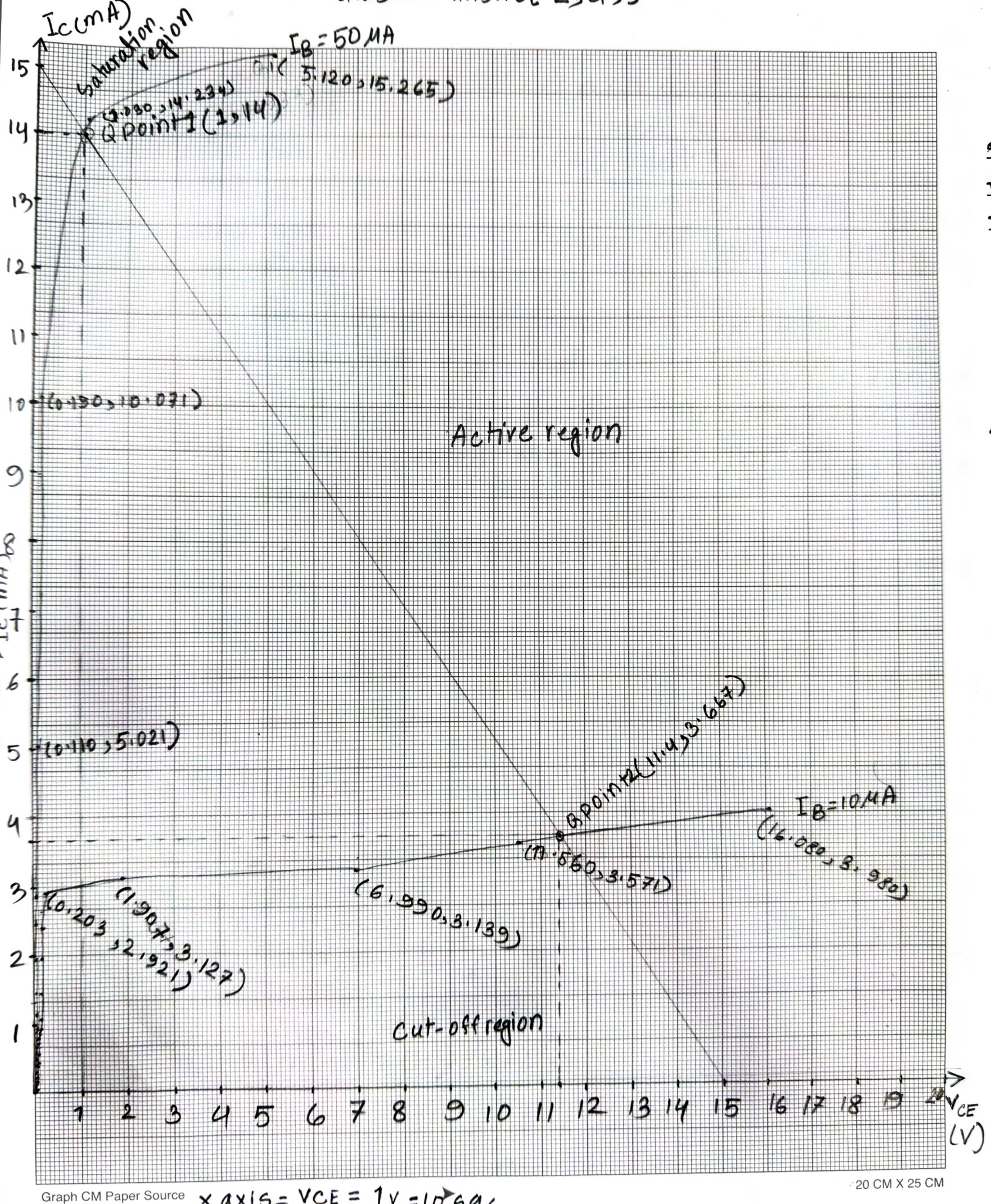
Graph CM Paper Source

20 CM X 25 CM

X axis = V_{BE} = 25 sq = 0.1 V

Y axis = I_B 30 sq = 2 MA

Question answer 2,4,5



x axis = VCE = 1v = 10 sq

Y axis = Ic = 1mA = 15 sq

Result: For the input characteristics, first we built the circuit with proper connection and set V_{CE} on 1v (nearly it was 0.992v) and also we changed the value of V_{BB} multiple times to observe the values of V_{BE} and V_{RB}. We calculated I_B then. We repeated the process of taking 5V(4.998v) by varying Vcc. We got different values of V_{BE}, V_{RB} and I_B by varying V_{BB} multiple times.
For the output characteristics, we made $V_{RB}=1V$ by varying V_{BB}. That maked $I_B = 10\mu A$. Then we measured V_{CE}, V_{RC}, I_C for different values of V_{CC}. Again we repeated the process for $V_{RB}=5v$ or $I_B = 50\mu A$.

One row calculation of I_B (input characteristics),

$V_{CC} = 1v$
$V_{BB} = 2v$
$V_{BE} = 0.628v$
$V_{RB} = 1.448v$

$$\therefore I_B = \frac{V_{RB}}{R_B} = \frac{1.448}{102.1} = 14.182\mu A$$

R_B (Practical) $= 102.1k\Omega$;

One row calculation of I_C (output characteristics),

$V_{RB} = 1V$
$V_{CC} = 2V$
$V_{CE} = 0.133V$
$V_{RC} = 1.939v$

$$I_C = \frac{V_{RC}}{R_C} = \frac{1.939}{0.98} = 1.979 mA$$

R_C (practical) $= 0.98k\Omega$;

Question Answer

Ans:1 Graph is attached with the report.

Ans:2 Graph is attached and different region of operation is also mentioned.

Ans:3 For active region,

When, $I_B = 10\mu A$,

$V_{CC} = 10\,V$,

$V_{CE} = \cancel{3.077V}$, $6.990V$,

$V_{RC} = 3.077V$

$I_C = 3.139mA$

$\therefore \beta = I_C/I_B = \frac{3.139}{10\times10^{-3}}$

$= 313.9$

Again, $V_{CC} = 5V$

$V_{CE} = 1.907V$

$V_{RC} = 3.064V$

$I_C = 3.127mA$

$\therefore \beta = \frac{3.127}{10\times10^{-3}}$

$= 312.7$

When, $I_B = 50\mu A$,

$V_{CC} = 10V$,

$V_{CE} = 0.190\,V$

$V_{RC} = 9.870\,V$

$I_C = 10.071\,mA$

$\beta = I_C/I_B = \frac{10.071}{50\times10^{-3}} = 201.42$

Again,

$V_{CC} = 5V$

$V_{CE} = 0.110\,V$

$V_{RC} = 4.921V$

$I_C = 5.021mA$

$\beta = \frac{5.021}{50\times10^{-3}}$

$= 100.42$

For different values of I_B and I_C we get different β

values in active region.

Ans:4 When $I_C = 0mA$,

$\Rightarrow V_{CE} = V_{CC} - I_C R_C$

$\Rightarrow V_{CE} = V_{CC} - 0$

$\Rightarrow V_{CE} = 15V$

When $V_{CE} = 0V$

$\Rightarrow 0 = V_{CC} - I_C R_C$

$\Rightarrow I_C = \frac{V_{CC}}{R_C} = \frac{15}{1} mA = 15mA$

By connecting points (15,0) & (0,15) we get load line, which intersects the curve of I_C vs V_{CE} for $I_B = 10\mu A$ and $I_B = 50\mu A$. So, we get two Q points. They are:

$I_B = 50\mu A$ curve, $Q_1(1,14)$

$I_B = 10\mu A$ curve, $Q_2(11.4, 3.667)$

Ans:5 From the graph, we observe that, Q point cuts the active region of operation.

Ans:6 Yes, the transistor successfully performed as an amplifier. Because, Q point of the biased point is within the active region of the transistor. Small change or variation in the base current or voltage creates a proportional but much larger change in the collector current, amplifying the signal by operating in the active region.

Discussion: This experiment successfully demonstrated the input and output behavior of a BJT in common-emitter mode. The input curve matched the expected diode like exponential relationship, confirming that the $V_{BE} \approx 0.7V$ is required for noticeable base current flow. The output characteristics clearly showed the three regions:

Cutoff region:- No base current, transistor off.

Active region:- A nearly constant I_c for increasing VCE, showing that the transistor can amplify signals here.

Saturation region:- Low VCE, I_c no longer depends on ~~#~~ I_B, indicating the transistor is fully on.

The current gain $\beta = I_c/I_B$ was observed to be consistant within the active region, proving that BJT operates as a current amplifier. Small changes in I_B produced significant changes in I_c, showcasing the transistor's amplification capability. Though we faced some difficulties to built the circuit but with the help of respected lab instructor we were able to built it afterward and ~~tak~~ took all necessary values of input-output characteristics.

Overall, the experiment confirms the principle of transistor action and reinforces the importance of selecting the correct Q-point for linear amplification. The CE configuration is highly preferred in practical circuits because it provides high voltage gain, moderate current gain, and good input-output isolation.

Output Characteristics:

1. Connect the circuit as shown in the circuit diagram.
2. By varying $\mathbf{V_{BB}}$, make $\mathbf{V_{RB}}$ = 1V. This makes $\mathbf{I_B}$ = 10μA.
3. Varying $\mathbf{V_{CC}}$ gradually in steps, measure Collector-Emitter Voltage, $\mathbf{V_{CE}}$ and $\mathbf{V_{RC}}$. Calculate $\mathbf{I_C} = \frac{V_{RC}}{R_C}$. Fill up Table-2.
4. Repeat above procedure (step 3) for $\mathbf{I_B}$ = 50μA [V_{RB} = 5V]

Data Collection:

$R_B = 102.1\,k\Omega$

$R_C = 0.98\,k\Omega$

Signature of instructor:

11-11-2025

Experiment: 3,

Performed by Group# 1

Table 1: Input Characteristics of BJT

V_{CE} = 1 V (0.992)				V_{CE} = 5V (4.998)			
V_{BB} (Volts)	V_{BE} (Volts)	V_{RB} (Volts)	$I_B = \frac{V_{RB}}{R_B}$ (μA)	V_{BB} (Volts)	V_{BE} (Volts)	V_{RB} (Volts)	$I_B = \frac{V_{RB}}{R_B}$ (μA)
0.1	0.091	0.002	0.020	0.1	0.100	0.004	0.039
0.2	0.214	0.004	0.039	0.2	0.198	0.009	0.088
0.3	0.310	0.009	0.088	0.3	0.290	0.011	0.108
0.4	0.377	0.017	0.167	0.4	0.381	0.024	0.235
0.5	0.506	0.038	0.372	0.5	0.468	0.060	0.588
0.6	0.521	0.088	0.862	0.6	0.523	0.100	0.979
0.7	0.581	0.186	1.822	0.7	0.560	0.143	1.401
0.8	0.589	0.226	2.214	0.8	0.595	0.254	2.488
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2.0	0.628	1.448	14.182	2.0	0.659	1.416	13.869

Table 2: Output Characteristics of BJT

(1.036V) (5.022V)

	$I_B = 10\mu A$ or V_{RB}=1V			$I_B = 50\mu A$ or V_{RB}=5V		
V_{CC} √	V_{CE} (Volts)	V_{RC} (Volts)	$I_C = \frac{V_{RC}}{RC}$ (mA)	V_{CE} (Volts)	V_{RC} (Volts)	$I_C = \frac{V_{RC}}{RC}$ (mA)
0.1	0.020	0.070	0.071	0.017	0.110	0.112
0.2	0.035	0.148	0.151	0.029	0.239	0.244
0.3	0.048	0.243	0.248	0.032	0.310	0.316
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3.0	0.203	2.863	2.921	0.080	2.900	2.959
5.0	1.907	3.064	3.127	0.110	4.921	5.021
10.0	6.990	3.077	3.139	0.190	9.870	10.071
15.0	11.560	3.500	3.571	1.030	13.955	14.234
20.0	16.080	3.980	4.061	5.120	19.960	15.265

Report:

1. Plot I_B vs. V_{BE} for different values of V_{CE}.
2. Plot I_C vs V_{CE} for different values of I_B. Show different regions of operations.
3. Find β for the each I_B [for active region only]
4. For V_{cc}= 15V, draw the load line and write the coordinates of the Q-point.
5. Which region of operation does your Q-point cut?
6. Explain whether your transistor performed as an amplifier or not.